Date |
Page |

Bayesian Assignment

Q1. Let W = Manchester wins
P = pub is packed

Given
$P(W) = 0.7$ $P(\text{not } W) = 0.3$
$P(P/W) = 0.9$
$P(P/\text{not } W) = 0.6$

$$P(W/P) = \frac{0.9 \times 0.7}{0.9 \times 0.7 + 0.6 \times 0.3} = \frac{0.63}{0.81} = 0.78$$

Ans 2 Let F = Nurse forgot
D = Death

$P(F) = 0.3$, $P(\text{not } F) = 0.7$
$P(D/F) = 0.8$, $P(D/\text{not } F) = 0.1$

$$P(F/D) = \frac{0.8 \times 0.3}{0.8 \times 0.3 + 0.1 \times 0.7} = \frac{0.24}{0.31} = 0.77$$

Ans 3) Let G = Gold, C = Coal, N = Neither
Positive test = +
$P(G) = 0.1$, $P(C) = 0.3$, $P(N) = 0.6$
$P(+|G) = 0.8$, $P(+|C) = 0.4$,
$P(+|N) = 0.2$

Date |
Page |

$$P(a|+) = \frac{0.8 \times 0.1}{0.8 \times 0.1 + 0.4 \times 0.3 + 0.2 \times 0.6}$$

$$= \frac{0.08}{0.32} = 0.25$$

Ans 4) Let M = Meningtis
$+$ = positive

4.1 $P(M) = 0.05$, $P(\text{not } M) = 0.95$

$P(+|M) = 0.95$, $P(+|\text{not } M) = 0.3$

$$P(M|+) = \frac{0.0475}{0.3325} = 0.143$$

4.2 (Second test negative)

$P(-|M) = 0.05$, $P(-|\text{not } M) = 0.7$

$$P(M|+,-) = \frac{0.05 \times 0.143}{0.05 \times 0.143 + 0.7 \times 0.857}$$

$$= 0.012$$

Date |
Page |

Ans 5) Let R = Rain
Negative Result = $-$

Given
$P(R) = 0.8$, $P(\text{No Rain}) = 0.2$

$P(-|R) = 0.25$, $P(-|\text{No Rain}) = 0.85$

$$P(R|-) = \frac{0.25 \times 0.8}{0.25 \times 0.8 + 0.85 \times 0.2}$$

$$= \frac{0.2}{0.37} = 0.54$$

Let C = chikungunya + = Positive

Ans 6) 6.1 $P(C) = 0.0001$ $P(\text{not } C) = 0.9999$

$P(+|C) = 0.99$, $P(+|\text{not } C) = 0.04$

$P(C|+) = 0.0025$

6.2 $P(C|+,+) = 0.058$

5.8% $\Rightarrow$ increases but ~~still~~ low

Ans 7)

$P(\text{churned} = \text{Yes}) = \frac{5}{10} = \frac{1}{2}$

$P(\text{churned} = \text{No}) = \frac{5}{10} = \frac{1}{2}$

$P(\text{handset} = \text{odd}|\text{Yes}) = \frac{2}{5}$

$P(\text{Time} > 2.5 / \text{Yes}) = \frac{2}{5}$

$P(\text{Age} \leq 55 / \text{Yes}) = \frac{4}{5}$

$P(\text{old}/\text{No}) = 0$

$P(T \geq 2.5 / \text{No}) = \frac{3}{5}$

$P(A \leq 55 / \text{No}) = \frac{1}{5}$

$L(\text{Yes}) = P(\text{old}/\text{Yes}) \cdot P(>2.5/\text{Yes}) \cdot P(\leq 55/\text{Yes}) \cdot P(\text{Yes})$

$= 0.064$

$L(\text{No}) = P(\text{old}/\text{No}) \cdot P(>2.5/\text{No}) \cdot P(\leq 55/\text{No}) \cdot P(\text{No})$

$= 0$

$P(\text{churning}) = \frac{0.064}{0.064} = 1$

1.2 use m-estimate

$$P = \frac{n_c + m \cdot p}{n + m}$$

$p \to$ prior estimate

$m \to$ equivalent sample size

$n \to$ total sample

Date |
Page |

Let $m = 3$, $p = 0.5$

$$P(\text{olap}|\text{No}) = \frac{0 + 3(0.5)}{5+3}$$

$$= 0.1875$$

if m small → p rely on observed data
→ leads to 0
large → p towards uniform dist^n, prevent extremes outcomes

Ans 8 $P(+) = \frac{4}{8} = 0.5$ & $P(-) = 0.5$

for +ve (+)

$$\mu_{x^+} = \frac{5+5+3+4}{4}$$

$$= 4.25$$

$$\sigma^2_{x^+} = 0.6875$$

$$\mu_{y^+} = \frac{6+7+6+5}{4}$$

$$= 6$$

$$\sigma^2_{y^+} = 0.5$$

Date |
Page |

for negative (-)

$$\mu_{x-} = \frac{1+3+2+8}{4}$$

$$= 3.5$$

$$\mu_{y-} = \frac{7+4+6+1}{4}$$

$$= 4.5$$

$$\sigma^2_{y-} = 5.25$$

$$P(x/\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$P(7|+)$ ~~0.01097~~ $= 0.00197$

$P(4|+) = 0.0103$

$P(x,y|+) = 0.00197 \times 0.0103 = 0.0000203$

$P(+|x,y) = 0.5 \times 0.0000203$

$= 0.00001015$

Date |
Page |

for negative (−)

$= P(7|-) = 0.0636$

$P(4|-) = 0.170$

$P(x, y|-) = 0.0636 \times 0.170 = 0.0108$

$P(-|x, y) \propto 0.5 \times 0.0108 = 0.0054$

(7, 4) belongs to class (−)

Q9. 9.1 $\mu_1 = 2.3$, $\sigma_1 = 1.8$

$\mu_2 = 6.8$, $\sigma_2 = 3.2$

$P(C_1) = 0.5$, $P(C_2) = 0.5$

$\log L = \log(0.133) + \log(0.122) + \log(0.114)$
$+ \log(0.110) + \log(0.731)$

$\log L = -10.53$

9.2. yes using EM Algorithm

E-step = Assign probabilities of each point to each Gaussian

M-step: update mean and variance